

# OMANSHU BHATT

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Portfolio: <https://omanshu-bhatt-portfolio.vercel.app/> Github: <https://github.com/Omanshu09> LinkedIn: <https://www.linkedin.com/in/omanshubhatt/>

## PROFESSIONAL SUMMARY

Software Developer specializing in AI/ML and full-stack development. Builds end-to-end systems — from RAG-based retrieval and LLM reasoning pipelines to real-time collaborative applications. Comfortable shipping production-grade code across the stack.

## TECHNICAL SKILLS

**Languages:** Python, Go, Java, JavaScript, C

**Web & Frameworks:** React.js, FastAPI, Node.js, Django, HTML, CSS, Typescript

**Data & Databases:** Elasticsearch, PostgreSQL, Redis, Neon

**AI / ML:** PyTorch, LangChain, Generative AI, RAG, Vector Databases

**Cloud, DevOps & Systems:** Docker, Kubernetes, Kafka, CI/CD (Jenkins, GitHub Actions)

**Distributed & Realtime Systems:** Yjs, CRDTs, Ably, IndexedDB, REST APIs, Distributed State Synchronization

**Core CS:** Data Structures & Algorithms, Operating Systems, Database Management Systems, Object-Oriented Programming

## FEATURED PROJECTS

### Web Research & Evidence Engine

- Built a research-driven web evidence search engine for query processing, multi-source retrieval, content extraction, relevance filtering, ranking and agreement/contradiction detection, generating structured source-backed responses through LLM synthesis.
  - Tech:** Web Search, Information Retrieval, Evidence Extraction, RAG, LLM, FastAPI, React, Docker
  - Benchmarked across 100+ research queries, achieving ~90% relevant source retrieval with sub-10s median response time.
  - Independently architected and competitively presented at the Agents for Humans Hackathon | Amazon/AWS
- Code: <https://github.com/Omanshu09/Search-Engine> Live: <https://search-engine-frontend-xi.vercel.app/>

### Real-Time Collaborative Engineering Workspace

- Built a real-time collaborative workspace with multi-user CRDT-based Notepad editing, near-instant synchronization, independent Python execution, shared output, and automatic state recovery during connectivity loss.
  - Tech:** React, TypeScript, Vite, Node.js, FastAPI, Express, Yjs, CRDT, Ably, IndexedDB, PostgreSQL, Neon, E2B, REST APIs, Vercel, Render
  - Instant collaborative synchronization during multi user editing, with offline local state retention and isolated Python execution
  - Independently architected and competitively presented at GatewayHacks 2026 | Devpost
- Code: <https://github.com/Omanshu09/Nexus-deployment> Live: <https://nexus-deployment-frontend.vercel.app/>

### Engineering Intelligence Platform

- Built a browser based intelligence platform that analyzes public GitHub repositories to uncover code structure, symbols, imports, dependencies and architectural relationships, presenting the results through an interactive repository explorer and dependency graph.
  - Tech:** JavaScript, TypeScript, React, Web Workers, Static Analysis, GitHub API, Dependency Graphs, JSON, Vercel Serverless Functions
  - Validated across diverse GitHub repositories for reliable code intelligence, dependency analysis, and architectural insights.
- Code: <https://github.com/Omanshu09/FORGE> Live: <https://forge-omanshu.vercel.app/>

## EXPERIENCE

### Trainee Intern | Indian Oil Corporation Limited (IOCL) | July 2025 – August 2025

- Worked with operational data and database systems to analyze trends, identify recurring patterns, and support data-backed improvements in engineering processes.
- Collaborated directly with IOCL engineers and technical teams, gaining hands-on experience in database management, enterprise engineering workflows, and quality-focused problem solving.

### Virtual Intern — AI/ML & Robotics | Kodacy | July 2024 – August 2024

- Applied Python to AI/ML and data-analysis tasks, working with datasets to extract patterns, evaluate results, and build practical solutions.
- Developed Python-based robotics simulation prototypes, gaining exposure to algorithmic problem solving, automation and intelligent systems.

## CERTIFICATIONS

- Oracle — Agentic AI Certified Foundations Associate
- TCS iON Career Edge - Young professional
- NASA — Hyperspectral Data for Land and Coastal Systems
- Deloitte — Cyber Job Simulation
- Commonwealth Bank — Software Engineering Job Simulation
- University of Oxford — AI, Justice and rule of law
- BCGX — GenAI Job Simulation
- Cisco — Data Science

## EDUCATION

- Govind Ballabh Pant University of Agriculture and Technology, Pantnagar | 2022 – 2026 | CGPA: 7.2**  
Bachelor of Technology (B.Tech) - Computer Engineering
- Beersheba Senior Secondary School, Almora**  
Class XII | 2021 – 2022 | 90.4%  
Class X | 2019 – 2020 | 95.3